

Inventory Service Design Document

Purpose: Source of truth for seat availability during booking. Focuses on concurrency, seat holding, confirmation, release and expiration.

Responsibilities

Maintain seat availability; Hold seats; Confirm seats; Release seats; Handle hold expiration.

Owned Entities

Seat, ShowSeat, SeatHold

Published Events

SeatHeld, SeatHoldFailed, SeatReleased, SeatConfirmed

Consumed Events

ScreenCreated, ScreenDeleted, ShowCreated, ShowCancelled

Architecture

Catalog publishes ShowCreated. Inventory creates ShowSeat records for the show. Booking requests interact only with Inventory for availability. Booking Service never accesses Inventory database directly.

ER Diagram (Text)

Seat

seat_id (PK)

screen_id

row_number

seat_number

seat_type

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ShowSeat

show_id

seat_id (FK)

status

version

hold_id

hold_expiry

SeatHold

hold_id (PK)

booking_id

user_id

expires_at
status

Database Schema

Seat: seat_id(UUID), screen_id(UUID), row_number, seat_number, seat_type, created_at, updated_at

ShowSeat: show_id(UUID), seat_id(UUID), status, version(BIGINT), hold_id(UUID), hold_expiry(TIMESTAMP), price, created_at, updated_at

SeatHold: hold_id(UUID), booking_id(UUID), user_id(UUID), created_at, expires_at, status

Indexes

Seat(screen_id); ShowSeat(show_id,status); ShowSeat(show_id,seat_id UNIQUE);
SeatHold(expires_at); SeatHold(booking_id)

Concurrency

Use optimistic locking via version on ShowSeat. During hold, update succeeds only if version matches. Background scheduler releases expired holds.

Business Flow

ShowCreated -> Create ShowSeat rows -> User selects seats -> Hold -> Payment -> Confirm OR Release on failure/expiry.